Adrenal & Retroperitoneum Rapid Revision

Campbell-Walsh-Wein · Chapters 140, 145–147

For MCh Urology theory & viva · NEET-SS / competitive MCQs

How to use (total ≈ 2h 09min even at a slow, careful pace)

- Each chapter reads top-to-bottom in the order you would write an essay answer: definition → pathology → diagnosis → management → complications → follow-up. The same material serves theory and viva.
- Boxed **ALGORITHM** and shaded **KEY POINT** panels carry the reasoning examiners ask about ("why do we…?").
- **MCQ one-liners** end each chapter: re-read these the day before the paper.
- The last section puts the big treatment choices side by side for essay questions.

Conventions: **underlined bold** = number to memorise · MCQ = exam trivia only · → *Ch 147* = cross-reference · † = not in the source chapter notes (added because it is essential in practice). Abbreviations are spelled out at first use in each chapter.

Chapter	Topic	Slow-reader time
145	Surgical & Radiographic Anatomy of the Adrenals	9 min
146	Adrenal Disorders: Pathophysiology, Evaluation & Medical Management	63 min
147	Surgery of the Adrenal Glands	18 min
140	Retroperitoneal Tumours	34 min
★	Putting it together: adrenal and retroperitoneal decisions	5 min
	Total (8966 words, at 70 words/min)	**2h 09min**

145 Surgical and Radiographic Anatomy of the Adrenals

Reading time ≈ 9 min (slow pace)

1 · EMBRYOLOGY

- **Cortex**: coelomic epithelium (mesoderm), weeks 3–4 → **adrenogonadal ridge** (cephalic part → cortex; caudal → gonad). **Medulla**: **neural crest** cells migrate in at weeks 8–10.
- At birth gland is **2×** adult weight; **fetal zone involutes in year 1**, replaced by definitive zone (ZG, ZF) under adrenocorticotropic hormone (ACTH). Zona reticularis (ZR) reaches adult volume at 6–8 y.
- **Renal agenesis**: adrenal still orthotopic, may be **discoid**.
- **Adrenal rests**: ≈1% adults (near coeliac axis); along gonadal descent in 7.5–15% neonates (regress). In congenital adrenal hyperplasia (CAH) → **testicular masses**.

2 · ZONES AND HORMONES

Cortex ≈ 90% of mass, medulla ≈ 10%. †Mnemonic outside → in: **GFR = salt, sugar, sex**.

Zone	Share	Cells	Product
Glomerulosa (outer)	~15%	Small, clustered, sparse lipid	Aldosterone †(renin–angiotensin, K^+ driven)
Fasciculata	**~80%**	Lipid-laden "clear cells" in straight cords	Cortisol
Reticularis (inner)	5–7%	Anastomosing cords	Dehydroepiandrosterone sulphate (DHEAS)
Medulla	~10%	Chromaffin cells	Epinephrine, norepinephrine

- Medulla = **modified sympathetic ganglion**: preganglionic splanchnic fibres pass through coeliac ganglion **without synapse** to chromaffin cells (postganglionic neurones without axons).

3 · RELATIONS

Inside Gerota fascia, 11th–12th rib level; ≈5 g; both on diaphragmatic crus, psoas behind.

	Right	Left
Shape / level	Triangular, higher, directly above upper pole	Crescentic, lower (to L1), medial to upper pole
Anterior	Liver; duodenum (anteromedial)	Splenic vessels, pancreatic body
Medial	**IVC** (inferior vena cava; gland is retrocaval)	Aorta
Exposure	Liver up, IVC medial, duodenum/kidney down	Pancreas + spleen up, splenic flexure medial

4 · BLOOD SUPPLY

Artery	Usual origin
Superior	Inferior phrenic artery
Middle	Aorta
Inferior	Renal artery

- Each source fans into 10–50 capsular twigs → **many arteries in, one central vein out**.
- Medulla perfused in series: cortical sinusoids drain into it, so high-cortisol blood drives **epinephrine synthesis**.
- **Right vein**: short, into **posterior IVC** → avulsion risk. **Left vein**: longer, into cranial **left renal vein**, often joined by inferior phrenic vein.

> **KEY POINT** Right side is riskier: high, retrocaval, short vein — little working length; avulsion tears the cava.

5 · IMAGING APPEARANCE

- **Computed tomography (CT)**: best spatial resolution; inverted **V/Y**, limbs 3–6 mm (thinner than crus). Right posterior to IVC; left anteromedial to upper pole, behind splenic vessels. Unenhanced **≤10 HU** (Hounsfield units) = lipid-rich tissue. Morphology only.
- **Magnetic resonance imaging (MRI)**: best contrast; coronal/sagittal confirm adrenal origin; no radiation. T1 slightly below liver; on **T2 resembles fat** (intracellular lipid).
- **Ultrasound (US)**: hypoechoic; role = solid vs cystic.
- **Angiography**: adrenal venous sampling (AVS) = only **functional** study; cavography for intraluminal tumour; arteriography rarely needed.

6 · MCQ ONE-LINERS

1. Cortex from coelomic mesoderm; medulla from neural crest.
2. Adrenal and gonad share the adrenogonadal ridge.
3. Renal agenesis: adrenal orthotopic, often discoid.
4. Neonatal adrenal is twice adult weight; fetal zone involutes in year 1.
5. Largest zone: fasciculata (~80%) — cortisol.
6. Glomerulosa → aldosterone; reticularis → DHEAS.
7. Testicular adrenal rests present in CAH.
8. Preganglionic fibres reach medulla without synapsing.
9. Arteries: superior (inferior phrenic), middle (aorta), inferior (renal).
0. Right adrenal vein: short, into posterior IVC.
11. Left adrenal vein: into left renal vein, joined by inferior phrenic.
12. Cortical venous cortisol aids medullary epinephrine synthesis.
13. Normal CT: inverted V/Y, limbs thinner than crus.
14. ≤10 HU unenhanced = lipid-rich adrenal tissue.
15. On T2 MRI the normal gland resembles fat.
16. US principal role: solid vs cystic mass.
17. Only angiographic test giving function: adrenal venous sampling.
18. Left gland lies behind pancreas and splenic vessels.
19. Adrenal weight ≈5 g, equal in both sexes.
20. Zona reticularis expands at 6–8 years.

146 Pathophysiology, Evaluation and Medical Management of Adrenal Disorders

Reading time ≈ 63 min (slow pace)

1 · APPLIED ANATOMY AND PHYSIOLOGY

- **Two organs in one capsule**: cortex from intermediate mesoderm (endocrine, steroids); medulla from neural crest (neurocrine, catecholamines). 4–5 g; right triangular, left crescentic.
- **Veins**: right short, straight into inferior vena cava (IVC); left long, joined by inferior phrenic vein → left renal vein. Anomalous venous drainage in up to **9%** – matters for adrenal vein sampling (AVS). Arterial supply triple (inferior phrenic, aorta, renal) and overlapping → partial adrenalectomy rarely infarcts the remnant.

Zone	Product	Main regulator
Glomerulosa	Aldosterone (CYP11B2)	Angiotensin II and serum K^+; adrenocorticotropic hormone (ACTH) weak
Fasciculata	Cortisol (10–20 mg/day)	ACTH (circadian; peak early morning, nadir ≈11 pm)
Reticularis	Dehydroepiandrosterone (DHEA), DHEA-sulphate (DHEA-S), androstenedione	ACTH
Medulla	Epinephrine 80%, norepinephrine 19%, dopamine 1%	Preganglionic sympathetic fibres (T11–L2)

- Because the glomerulosa obeys angiotensin II and K^+ rather than ACTH, **pituitary failure (or exogenous steroid suppression) spares aldosterone**; ACTH is trophic to the inner zones, which atrophy without it.
- Aldosterone reabsorbs Na^+ (with water) and secretes K^+/H^+ in the distal nephron → volume rises, **not serum sodium**.
- Phenylethanolamine-N-methyltransferase (PNMT, norepinephrine → epinephrine) is almost unique to the medulla and induced by cortical cortisol – hence mainly adrenal tumours make epinephrine.
- Catechol-O-methyltransferase (COMT) inside chromaffin cells converts norepinephrine → normetanephrine and epinephrine → metanephrine continuously; >90% of circulating metanephrine is adrenal → **metanephrines are the diagnostic currency** of phaeochromocytoma. Vanillylmandelic acid (VMA) is the hepatic end product (monoamine oxidase).
- Receptor map: α1 vasoconstriction (hypertension, pallor), β1 (palpitations), α2 insulin suppression (hyperglycaemia → post-resection rebound hypoglycaemia).

2 · EVALUATION OF AN ADRENAL MASS – PRINCIPLES

> **KEY POINT** Two questions: **(1) Is it malignant?** – answered by imaging. **(2) Is it hormonally active?** – answered by biochemistry. Either = treat (usually resect). No imaging feature separates functional from non-functional adenoma, so biochemistry is never omitted.

Category	Indications for adrenalectomy (Box 146.1)
Functional	Autonomous cortisol secretion; phaeochromocytoma; lateralised aldosterone excess
Oncological	>4 cm (except myelolipoma); suspicious imaging (lipid-poor, heterogeneous, irregular, infiltrating); growth >1 cm on follow-up; isolated metastasis (multidisciplinary)
Myelolipoma	Very large and/or symptomatic
At nephrectomy for renal cell carcinoma (RCC)	Adrenal abnormal or not visualised; thrombus to adrenal vein level

Category	Indications for adrenalectomy (Box 146.1)
Bilateral	Failed pituitary surgery for Cushing disease; selected ectopic ACTH; AIMAH; PPNAD (§7)

3 · IMAGING

Lipid assay, then perfusion assay

- **Unenhanced computed tomography (CT)** – single best test. **≤10 Hounsfield units (HU)** = intracellular lipid = adenoma (in a patient without known malignancy); diagnoses >70% of adenomas. ≤0 HU is 100% specific but insensitive. Measure density yourself if the report omits it.
- Limits: up to **7%** of metastases are <10 HU; rare low-density phaeochromocytomas exist → biochemistry for every lesion.
- **Chemical-shift magnetic resonance imaging (MRI)**: signal drop-out on opposed-phase vs in-phase = intracellular lipid; broadly equivalent to unenhanced CT. Gadolinium washout is not used (signal not dose-linear).
- **CT washout** (gold standard for lipid-poor lesions, ≈30% of adenomas): adenomas wash out contrast fast, whether lipid-rich or lipid-poor.

	Absolute percentage washout (APW)	Relative percentage washout (RPW)
Phases	Unenhanced, enhanced (60 s), delayed (15 min)	Enhanced + delayed
Formula	(E − D) ÷ (E − U) × 100	(E − D) ÷ E × 100
Adenoma if	**>60%**	**>40%**

- **Pitfalls**: RCC and hepatocellular carcinoma metastases can wash out like adenomas; adrenocortical carcinoma (ACC) and phaeochromocytoma classically do not. Insist on the 15-minute delay – a badly done study generates needless follow-up.

ALGORITHM
Mass → unenhanced CT ≤10 HU or MRI drop-out → adenoma │ >10 HU → 15-min washout CT → APW >60% / RPW >40% = adenoma; below = adenoma very unlikely → manage as indeterminate

Size, growth and other features

Size	Malignancy (large unselected series)	Surgical series
<4 cm	0.1%	≈2–5% ACC
4–6 cm	2.4%	≈6–10%
>6 cm	19.5%	≈25%

- **4 cm** = consensus threshold to consider resection in fit patients (sensitive, poorly specific); 6 cm is more specific. CT underestimates pathological size.
- **Benign imaging overrides size**: no <10 HU lesion was malignant in the largest series. Age modifies: be cautious with small lesions in the young; observe large benign-looking lesions in old, comorbid patients.
- Suspicious: heterogeneity, irregular margins, enhancing rim, invasion. **Macroscopic fat** (<−30 HU) = myelolipoma. **Calcification is not reassuring** (up to 30% of ACC).

Follow-up	Classic	European Society of Endocrinology–European Network for the Study of Adrenal Tumors (ESE–ENSAT)
Re-imaging	6, 12 (± 24) months	None if <4 cm, benign imaging and silent; otherwise 6–12 months
Concerning growth	≥1 cm → consider resection	>20% diameter increase and ≥5 mm

Functional imaging

Modality	Best use	Weakness
^{18}F-fluorodeoxyglucose positron emission tomography (FDG PET)	Malignancy (ACC, metastasis) when CT inconclusive; staging phaeochromocytoma/paraganglioma (PPGL) – best for metastases, succinate dehydrogenase subunit B (SDHB) and von Hippel–Lindau (VHL)	Poor in multiple endocrine neoplasia type 2 (MEN-2); RCC/hepatoma metastases
^{68}Ga-DOTATATE PET/CT	PPGL; replacing FDG at some centres	—
^{123}I-metaiodobenzylguanidine (MIBG)	Extra-adrenal, recurrent, metastatic PPGL; biochemistry positive but CT negative; **best in MEN-2**; isolated normetanephrine excess (? extra-adrenal)	Cold in VHL/SDHB and dedifferentiated metastases

- Phaeochromocytoma >5 cm: MIBG or FDG PET before surgery to exclude metastases. Iodinated contrast does **not** precipitate crisis.

4 · HORMONAL WORK-UP (ALL LESIONS >1 CM)

> **KEY POINT** Every incidentaloma: **1-mg dexamethasone suppression test (DST) + metanephrines**. Add **aldosterone-to-renin ratio (ARR)** if hypertension or unexplained hypokalaemia. >10% of incidentalomas are functional, yet 75–80% are inadequately evaluated. **Never biopsy, sample or excise before excluding phaeochromocytoma** – manipulation can provoke a fatal catecholamine surge. Canadian Urological Association (CUA) guidance allows omitting metanephrines in asymptomatic <10 HU lesions.

Incidentalomas: cortisol adenoma 5%, phaeochromocytoma 5%, aldosteronoma 1%, ACC 5%, metastasis 2.5% → **≈18%** potentially surgical.

Cortisol

Test	Method / threshold	Pitfalls
Overnight 1-mg low-dose DST (preferred for adrenal masses)	1 mg at 11 pm–midnight, cortisol 8–9 am. **≤1.8 µg/dL (50 nmol/L)** excludes; 1.8–5.0 = possible autonomous secretion; **>5.0 µg/dL (140 nmol/L)** >95% specific	Oral oestrogens (↑ cortisol-binding globulin) → false positive; CYP3A4 inducers (phenytoin, carbamazepine, rifampicin) → false positive; inhibitors (itraconazole, ritonavir, diltiazem) → false negative; unaffected by renal function
Late-night salivary cortisol (LNSC) ×2	Bedtime saliva; loss of nocturnal nadir	Depression, shift work; preferred in epilepsy
24-h urinary free cortisol (UFC) ×2	Integrated free cortisol; check creatinine	Insensitive for mild excess; unreliable if glomerular filtration rate (GFR) <60 mL/min; test of choice in pregnancy

- "Low-dose" 1 mg is 3–4× physiological and is not detected by cortisol assays, so normals suppress; the 1.8 cut-off is used because the 5.0 cut-off misses up to 18% of Cushing syndrome.
- Exclude exogenous steroids first (creams, inhalers, nasal sprays). Pseudo-Cushing states (depression, alcohol, obesity, stress) also raise cortisol.
- **Terminology**: Cushing syndrome = clinical hypercortisolism; Cushing disease = pituitary ACTH adenoma; **mild autonomous cortisol secretion (MACS)** = autonomy without overt signs (replaces "subclinical Cushing" – it raises cardiovascular mortality, diabetes, bone disease).

Catecholamines

	Plasma free metanephrines	24-h urinary fractionated metanephrines
Performance	Highest sensitivity (97–99%) – best rule-out	Sensitivity >97% by tandem mass spectrometry

	Plasma free metanephrines	24-h urinary fractionated metanephrines
Conditions	**Supine after 20 min rest**, fasting, no caffeine	Complete collection, check creatinine; needs normal renal function
Highly indicative	Normetanephrine >2.2 nmol/L and/or metanephrine >1.2 nmol/L (≈4× upper limit)	—

- Liquid chromatography assays preferred. **False positives**: paracetamol, mesalazine, sulfasalazine (stop ≥5 days), tricyclics, **phenoxybenzamine**, severe illness, seated sampling.
- Equivocal rise → repeat supine off β-blockers, urinary test, chromogranin A (falls in specificity when GFR <80 mL/min), or **clonidine suppression** (α2-agonist suppresses neural, not tumour, norepinephrine).

ARR – §6. Sex steroids only if ACC suspected (§10).

5 · BIOPSY AND INCIDENTALOMA MANAGEMENT

Biopsy – rarely

- Cytology cannot separate adenoma from carcinoma; biopsy answers only "is this a metastasis?". ESE–ENSAT: **no biopsy without a history of malignancy**.
- Complications 3–12% (bleeding, pneumothorax); haemorrhage hampers later laparoscopy; seeding in ACC.

ALGORITHM
Imaging characterises it? → no biopsy | functional? → no biopsy | ACC suspected? → no biopsy | will result change management (e.g., known primary) and justify risk? → biopsy, **after excluding phaeochromocytoma**

Incidentaloma and non-functional adenoma

- Incidentaloma = adrenal mass >1 cm on imaging done for another reason (not cancer staging); ≈5% of scans, rising with age.
- Adenoma: >85% of incidental neoplasms; ≈93% inactive.
- **Inactive adenoma**: no treatment. Repeat hormones – none (European Society of Endocrinology) vs annual (CUA, American Association of Clinical Endocrinology). Risk of new hormone excess <1–1.7%. No follow-up imaging if <10 HU and <4 cm; indeterminate unresected lesions – re-image 6, 12 ± 24 months.

6 · PRIMARY ALDOSTERONISM (PA)

- Commonest secondary hypertension: 3–13% of hypertensives. Conn (1955) – first aldosteronoma.
- **Primary**: autonomous aldosterone → renin suppressed. **Secondary**: high renin drives aldosterone.
- No hypernatraemia and no oedema – **mineralocorticoid escape** (pressure natriuresis, atrial natriuretic peptide) caps volume expansion. Most (63–91%) are **normokalaemic**; hypokalaemia in <40%.
- Target-organ damage exceeds essential hypertension: stroke ≈4×, myocardial infarction ≈6.5×, atrial fibrillation ≈12×; left ventricular hypertrophy, proteinuria.

Subtype	%	Surgically curable?
Bilateral (idiopathic) hyperplasia	60	No – mineralocorticoid receptor antagonist (MRA)
Aldosterone-producing adenoma	35	Yes
Unilateral hyperplasia	2	Yes
Carcinoma, ectopic (ovarian), familial (FH)	<1 each	Carcinoma/ectopic yes; FH no

Familial	Genetics	Point
FH-I (glucocorticoid-remediable)	Autosomal dominant chimeric gene: CYP11B1 (ACTH-responsive) promoter + CYP11B2 coding → aldosterone under ACTH	Early severe hypertension, young stroke; treat with glucocorticoid ± MRA
FH-III	KCNJ5 (K^+ channel)	Very young, bilateral hyperplasia, severe hypokalaemia

Whom to screen (Endocrine Society 2016)

- Resistant hypertension (≥3 drugs incl. diuretic) or controlled on ≥4; BP ≥150/100 on three days; hypertension + hypokalaemia (spontaneous or diuretic); **incidentaloma + hypertension**; sleep apnoea; onset <20 y or stroke <50 y; family history of PA; disproportionate target-organ damage.

Screen – ARR

- Morning (8–10 am), seated: plasma aldosterone concentration (PAC, ng/dL) ÷ plasma renin activity (PRA, ng/mL/h). Positive: **ARR >20–30 with PAC >15–20 ng/dL**. Floor PRA at 0.2 to avoid spuriously high ratios; high ARR with low PAC suggests false positive.
- Correct hypokalaemia; avoid liquorice.

Drug	Effect on ARR	Action
Spironolactone, eplerenone, amiloride	Profound renin–angiotensin–aldosterone system (RAAS) change	Stop ≥4–6 weeks; substitute α1-blocker or long-acting calcium channel blocker
β-blockers	Renin ↓ → **false positive**	May continue (or stop)
ACE inhibitors / angiotensin receptor blockers	Renin ↑ → false negative	Continue; undetectable renin despite them is itself suspicious
Calcium channel blockers	Neutral	Continue

RAAS = renin–angiotensin–aldosterone system; ACE = angiotensin-converting enzyme.

Confirm (aldosterone fails to suppress with sodium)

Test	Protocol	Positive
Oral sodium loading	3 days high salt; 24-h urine Na >200 mmol	Urine aldosterone >12 µg/day
IV saline infusion	2 L 0.9% saline over 4 h, recumbent	PAC >5 ng/dL (>10 suggests adenoma)
Captopril challenge	25–50 mg	PAC stays >15 ng/dL – when salt load contraindicated (heart/renal failure)

- **Skip confirmation** if spontaneous hypokalaemia + undetectable renin + PAC >20 ng/dL. Only 50–87% of positive screens confirm.

Subtyping – lateralisation decides surgery, not CT

- Only surgical candidates are subtyped. CT for all (excludes ACC), but ≈20% of aldosteronomas are <1 cm and non-functioning adenomas are common → CT alone misdirects ≈25%.
- **AVS** (sensitivity 95%, specificity 100% if done well): after correcting K^+, morning, recumbent; catheterise both adrenal veins + IVC; measure aldosterone and cortisol.
  - **Selectivity**: adrenal vein/IVC cortisol >1.1–5:1 (higher with cosyntropin) – proves the catheter is in the vein.
  - **Lateralisation**: (A/C dominant) ÷ (A/C non-dominant) **>2–4:1** → ipsilateral adrenalectomy. Cortisol correction adjusts for dilution.
- **AVS may be omitted**: age <40 with clear unilateral adenoma >1 cm and normal contralateral gland; suspected ACC. Genetic testing if onset <20 y or family history.
- Adjuncts if AVS fails: NP59 (iodocholesterol) scintigraphy, posture test (adenoma: no upright PAC rise).

ALGORITHM
Indication → ARR + → confirm → (family history/onset <20 → FH-I test) → surgical candidate? No → MRA | Yes → CT → age <40 + unilateral >1 cm nodule → adrenalectomy; otherwise → AVS → lateralised → laparoscopic adrenalectomy; not lateralised → MRA

Treatment

- **Unilateral** → laparoscopic adrenalectomy (open if ACC suspected). Biochemical cure ≈94%; BP improves in most but only 37–52% stop all drugs.Residual hypertension: older, male, obese, long duration.
- Post-op: check PAC/ARR; † stop K^+ supplements and MRA; high-salt diet and watch for **hyperkalaemia** for weeks (contralateral glomerulosa suppressed); BP settles over 6 months.
- **Bilateral/unfit** → MRA: spironolactone 25–50 mg → max 400 mg (gynaecomastia, erectile dysfunction, menstrual upset); eplerenone 25 → 100 mg (selective, fewer side effects). Add weight loss, low salt.

7 · CUSHING SYNDROME AND CORTISOL EXCESS

Group	Cause	Share
Exogenous	Iatrogenic glucocorticoid	Commonest overall
ACTH-dependent (80–85% of endogenous)	Pituitary – Cushing disease	≈70%
	Ectopic ACTH (bronchial carcinoid, small cell lung cancer, thymic, islet cell, medullary thyroid, phaeochromocytoma; occult in up to 1/3)	≈10–15%
	Ectopic corticotropin-releasing hormone (CRH)	<1%
ACTH-independent (adrenal)	Adenoma, ACC, AIMAH, PPNAD	≈10%

- Central obesity (90–100%) is least specific; proximal weakness, bruising, purple striae discriminate.
- **Urological relevance**: hypogonadotropic hypogonadism (low libido, erectile dysfunction, low testosterone with low gonadotropins) and **urolithiasis in up to 50%** – test cushingoid stone formers.

ALGORITHM
Exclude exogenous steroid → screen (1-mg DST / LNSC ×2 / UFC ×2) → abnormal → endocrinologist confirms → **morning ACTH**: low → adrenal (CT) | normal/high → pituitary MRI + **inferior petrosal sinus sampling after CRH** (central:peripheral gradient = Cushing disease; none = ectopic)

ACTH-dependent disease – the urologist's role

- **Cushing disease**: trans-sphenoidal resection cures 60–80% (macroadenoma <15%), relapse up to 25%; radiotherapy is slow.
- **Bilateral adrenalectomy**: after failed pituitary treatment or life-threatening hypercortisolism; also for unresectable/occult ectopic ACTH (primary resectable in only ≈10%). Rapid, near-certain cure without hypopituitarism, at the price of lifelong glucocorticoid + fludrocortisone, crisis risk, and **Nelson syndrome** (8–29%: enlarging pituitary adenoma → chiasmal compression, oculomotor palsies).

ACTH-independent disease

	ACTH-independent macronodular adrenal hyperplasia (AIMAH)	Primary pigmented nodular adrenocortical disease (PPNAD)
Glands	Massive, nodules up to 4 cm, >60 g each	Normal size; small black/brown nodules in atrophic cortex
Association	McCune–Albright	Carney complex (≈half): spotty pigmentation, Sertoli cell tumours
Clue	Looks like ACTH-driven hyperplasia – check ACTH	Paradoxical rise in UFC after dexamethasone
Treatment	Bilateral adrenalectomy; unilateral removal of the larger gland has worked in AIMAH	

Cortisol-secreting adenoma and MACS

- Overt Cushing → adrenalectomy. Before surgery: confirm low ACTH; screen for vertebral fractures, diabetes, hypertension.
- **MACS**: surgery controversial – individualise; consider if hypertension, obesity, dysglycaemia or osteopenia. Progression to overt Cushing is very unlikely.
- **Postoperative adrenal insufficiency is expected** – the contralateral gland is atrophic from ACTH suppression: ≈65% after MACS, ≈99.7% after overt Cushing. Deaths from hypocortisolism have occurred.
- Protocol: hydrocortisone 100 mg IV on the day of surgery → oral taper (hydrocortisone 25–50 mg/day). Alternative: dexamethasone 0.5 mg 6-hourly (not measured by assay) → early-morning cortisol <50 nmol/L confirms cure → switch to hydrocortisone. Maintenance 10–12 mg/m²/day in 2–3 doses.
- hypothalamic–pituitary–adrenal (HPA) axis recovers in median 6.5 (MACS) – 11 (Cushing) months; check morning cortisol 3-monthly, then ACTH stimulation. Cardiovascular risk persists.
- **Medical (bridge/unfit)**: metyrapone, ketoconazole, etomidate, mitotane, mifepristone. Exogenous Cushing: taper slowly; beware steroid withdrawal syndrome.

8 · PHAEOCHROMOCYTOMA AND PARAGANGLIOMA (PPGL)

- Catecholamine-secreting tumour of chromaffin cells; adrenal 75–90%; **paraganglioma** = extra-adrenal (organ of Zuckerkandl, bladder, para-aortic, head and neck – latter usually non-functional). ≈0.5% of hypertension; ≈5% of incidentalomas.

"Rule of 10s"	Contemporary reality
10% extra-adrenal	Up to 25%
10% familial	**≈1/3**; ≈24% of apparently sporadic carry germline mutations
10% malignant	Sporadic adrenal ≤5%; >1/3 of extra-adrenal; SDHB 30–50%
10% bilateral	Common in hereditary disease
10% paediatric	Children: up to 40% familial, >20% bilateral

> **KEY POINT** The rule of 10s is outdated – every patient with PPGL should be offered genetic counselling/testing (Endocrine Society 2014).

Clinical

- Triad: headache (60–90%), sweating, palpitations. **Sustained hypertension is commoner than paroxysmal** (30–50%); pallor (not flushing); hyperglycaemia; orthostatic hypotension. >20% asymptomatic.

Hereditary syndromes

Syndrome	Gene (inheritance)	Features	PPGL risk / malignancy
MEN-2A	RET 10q11.2 (AD)	Medullary thyroid cancer, hyperparathyroidism	50% / ≈3%; almost always adrenal, bilateral
MEN-2B	RET (AD)	Medullary thyroid cancer, mucosal neuromas, marfanoid	50% / ≈3%
VHL type 2	VHL 3p25 (AD)	RCC, haemangioblastomas, pancreatic cysts, epididymal cystadenoma	10–20% / ≈5%; normetanephrine only
Neurofibromatosis 1	NF1 17q11 (AD)	Neurofibromas, café-au-lait	1% / ≈11%
PGL-4	**SDHB** 1p36 (AD)	Abdominal/thoracic paragangliomas	≈20% / **30–50%**
PGL-1	SDHD 11q23 (**maternal imprinting** – paternally transmitted)	Head and neck, multifocal	≈20% / <3%

AD = autosomal dominant. VHL subtypes: 1 – no phaeochromocytoma; 2A – phaeo, no RCC; 2B – phaeo + RCC; 2C – phaeo only. VHL and SDHB patients are often already under urological surveillance for RCC.

Diagnosis – biochemistry first, then localise

- Metanephrines (§4) – even in a suspected solitary metastasis.
- CT: >10 HU (mean ≈35), vascular, no rapid washout. MRI: no drop-out; T2 "light bulb" sign neither sensitive nor specific.

Preoperative preparation – all with abnormal biochemistry, even if normotensive

Drug	Mechanism	Use
Phenoxybenzamine (option 1, first line)	Irreversible non-selective α-blockade – surges cannot overcome it	Start 7–14 days pre-op: 10 mg twice daily, ↑ every 2–3 days (≈1 mg/kg total); last dose night before. Causes post-resection hypotension and false-positive metanephrines
Doxazosin, prazosin, terazosin	Selective reversible α1	Fewer side effects, less post-op hypotension
Calcium channel blockers (amlodipine, nifedipine; option 2 or add-on)	Vascular smooth muscle relaxation	Sole therapy for mild cases; no reflex tachycardia or post-op hypotension
β1-blockers (atenolol, metoprolol)	Rate control	For tachycardia/arrhythmia **only after adequate α-blockade**
Metyrosine (α-methyl-p-tyrosine)	Inhibits tyrosine hydroxylase (rate-limiting)	250 mg 2–3×/day ↑ to 1.5–2 g; with α-blockade for large/metastatic tumours; sedation, depression, galactorrhoea, **extrapyramidal** effects

> **KEY POINT** **Why never β before α?** β-blockade removes β2-mediated arteriolar dilatation; circulating catecholamines then act unopposed on α1 receptors → hypertensive crisis. Hence use selective β1 agents.

- **Targets**: seated BP 120–130/80 mmHg; mild orthostatic hypotension (standing systolic >80–90 mmHg) = adequate blockade; † heart rate 60–70/min seated.
- **Volume repletion** – perhaps the most important step: catecholamine-constricted circulation is contracted. Liberal salt and fluid once blockade starts; admit night before, 1–2 L IV crystalloid.

Surgery and postoperative care

- Laparoscopic adrenalectomy standard; open if large/invasive. **MEN-2 and VHL** (low malignancy, high bilaterality): **cortical-sparing partial adrenalectomy** to avoid lifelong steroids – recurrence possible, replacement may still be needed.
- Post-op: **hypotension** (persistent phenoxybenzamine, sudden catecholamine withdrawal) → fluids ± vasopressors; **rebound hypoglycaemia** (insulin suppression via α2 lifted) → glucose monitoring; consider overnight intensive care.

Malignant PPGL and follow-up

- Malignancy is defined **only by metastases** (bone, lung, liver, nodes); histology and local invasion predict poorly. Risk: SDHB, extra-adrenal, >5 cm, dopamine-secreting.
- Treatment: metastasectomy, ablation, embolisation; **^{131}I-MIBG therapy** if avid; cyclophosphamide–vincristine–dacarbazine (CVD) – short responses. 5-year survival ≈50%.
- Follow-up: metanephrines at ≈6 months, then **annually lifelong** – recurrence up to 16% at 10 years, even >15 years later. Persistent post-op elevation → MIBG to seek metastases.

9 · NON-FUNCTIONAL BENIGN LESIONS

Lesion	Key features	Management
Myelolipoma	Mature fat + haematopoietic tissue; macroscopic fat on CT/MRI is diagnostic; calcification >25%; may exceed 10 cm	Exception to 4-cm rule. Still do hormonal work-up (coexistent adenoma). Excise if symptomatic or large (haemorrhage risk; bleeders average 10–14 cm, cut-offs 6–10 cm unvalidated); embolise acute bleed
Cysts	Endothelial 45%, pseudocyst 39% (post-haemorrhage; the type harbouring malignancy, ≈7%), epithelial 9%, parasitic (hydatid) 7%. Rim calcification; >30 HU = haemorrhage/debris	Endocrine work-up for all. Observe small, thin-walled, homogeneous, asymptomatic. **Excise if >5 cm, thick-walled, heterogeneous or symptomatic**; no simple decortication. Aspirate only after excluding phaeochromocytoma
Ganglioneuroma	Young; encases vessels without narrowing; punctate calcification	Diagnosed at resection; benign
Oncocytoma	Rare; large; ≈10% functional; no central scar	Resect; >30% reported malignant (unlike renal oncocytoma)

- Distinguish myelolipoma from liposarcoma (infiltrative).

10 · ADRENOCORTICAL CARCINOMA (ACC)

- 0.7–2 per million; bimodal (first decade; 4th–5th decades). Syndromes: **Li–Fraumeni (TP53)**, **Beckwith–Wiedemann (IGF2)**, Lynch, Carney, MEN-1, familial adenomatous polyposis.
- Functional in 50–79% of adults (90% of children): **cortisol** commonest (Cushing ± virilisation; cortisol secretion = poor prognosis), androgens next (virilisation in women), oestrogen (feminisation in a man = **carcinoma until proved otherwise**), aldosterone rare.
- Average 10–12 cm; >90% exceed 5 cm. CT: heterogeneous, necrosis, calcification, mean ≈39 HU, no washout. MRI for IVC thrombus. FDG-avid. Stage with CT chest–abdomen–pelvis (lung, liver); bone/brain if symptomatic.
- Full hormonal panel (cortisol, DHEA-S, testosterone, oestradiol, steroid precursors, metanephrines, ARR if hypertensive) – diagnosis, tumour markers for surveillance, and planning post-op steroid cover.
- **No preoperative biopsy** (seeding; cytology unhelpful) – biopsy only unresectable/metastatic disease before systemic therapy.

Pathology

- Steroidogenic factor 1 (SF1) = cortical origin. **Weiss criteria – ≥3 of 9 = malignant**: *cytology* (high nuclear grade, >5 mitoses/50 high-power fields, atypical mitoses); *structure* (low proportion of clear cells, diffuse architecture, necrosis); *invasion* (venous, sinusoidal, capsular). Mitotic rate grades ACC (low/high) and best predicts outcome. Use alternative criteria for paediatric and oncocytic tumours.

Staging

Stage	European Network for the Study of Adrenal Tumors (ENSAT)	Proportion at diagnosis
I	T1 (≤5 cm) N0 M0	3–4%
II	T2 (>5 cm) N0 M0	29–46%
III	T3 (periadrenal fat) or T4 (adjacent organ) or N1 or venous tumour thrombus	11–19%
IV	Distant metastasis only	39–49%

Difference from older AJCC/UICC staging: those placed T4 or T3N1 in stage IV; ENSAT keeps all non-metastatic disease in stage III. Overall 5-year survival 20–47%; stage IV <5%.

Management

- **Complete (R0) en bloc resection** is the only cure – open surgery for >6 cm or invasive tumours (higher peritoneal seeding and spillage with laparoscopy); en bloc removal of involved organs, regional lymphadenectomy, IVC thrombectomy ± bypass. Avoid capsule rupture – ACC is a disease of recurrence (60–80% after resection).
- Metastatic: debulk only if >90% can be removed; otherwise go straight to systemic therapy. Resect recurrences (low grade, long disease-free interval); ablation/embolisation if unresectable.
- **Radiotherapy**: bone/brain metastases; adjuvant to tumour bed for R1/Rx margins – cuts local recurrence, not survival.
- **Mitotane** (DDT derivative; adrenolytic – kills cortical cells and blocks steroidogenesis): needs glucocorticoid replacement († often higher than usual doses, because mitotane raises binding globulins and accelerates steroid clearance); † target serum level 14–20 mg/L. Toxicity: gastrointestinal, lethargy, gynaecomastia, hepatitis, hypothyroidism.
- † Adjuvant mitotane is reserved for high-risk resected disease (R1/Rx, Ki-67 >10%, stage III); ADJUVO showed no benefit in low-risk R0 disease.
- Advanced: **mitotane + etoposide, doxorubicin, cisplatin (M-EDP)** first line (FIRM-ACT). Hormone control: metyrapone, ketoconazole, etomidate, mifepristone.
- Surveillance: CT chest–abdomen–pelvis + markers 3-monthly × 2–3 y, then 6-monthly to 5 y.

Paediatric ACC: >90% functional (virilisation), test for Li–Fraumeni (TP53), better prognosis.

Neuroblastoma → *Ch 63*. Adrenal sarcomas: rare, wide excision.

11 · ADRENAL METASTASES

- With known malignancy, **>50%** of new adrenal lesions are metastases.
- Imaging: adenoma-like but lipid-poor (>10 HU, poor washout); RCC/hepatoma deposits may wash out → growth kinetics and biopsy decide. **Still exclude phaeochromocytoma** (functional rates are the same with or without prior cancer).
- Resect isolated metastasis in selected patients – best data in non-small cell lung cancer (level 1 evidence) and melanoma; RCC as part of cytoreduction.
- At nephrectomy, remove the ipsilateral adrenal only if it is abnormal on imaging or thrombus reaches the adrenal vein.

12 · ADRENAL INSUFFICIENCY

	Primary (Addison)	Secondary (pituitary/hypothalamic)
Causes	Autoimmune (West; anti-21-hydroxylase antibodies), **tuberculosis** (developing countries), HIV-related infection, bilateral haemorrhage, infiltration, **bilateral adrenalectomy**, drugs (ketoconazole, abiraterone)	**Exogenous glucocorticoid suppression** (commonest for surgeons), pituitary tumour/apoplexy, Sheehan, radiation, critical illness
Aldosterone	Lost	Preserved
Skin	Hyperpigmentation (high ACTH/POMC)	Pallor
Electrolytes	↓Na^+, ↑**K^+**, ↑creatinine; salt craving	↓Na^+ (SIADH), normal K^+
Replacement	Hydrocortisone + fludrocortisone	Hydrocortisone only

POMC = pro-opiomelanocortin; SIADH = syndrome of inappropriate antidiuretic hormone secretion (from cortisol deficiency, hence hyponatraemia in both).

- **Crisis**: fluid-refractory hypotension mimicking an acute abdomen (pain, vomiting, fever).
- **Urological at-risk settings**: adrenalectomy for cortisol-secreting tumour; bilateral (simultaneous or staged) adrenalectomy; **previous contralateral nephrectomy/adrenalectomy** before surgery on the remaining side – review old operative notes, pathology and imaging.
- Diagnosis: morning cortisol + ACTH; renin/aldosterone in primary; confirm with ACTH (cosyntropin) stimulation test.
- Maintenance: hydrocortisone in 2–3 doses, ½–⅔ on waking (circadian mimicry); fludrocortisone in primary only.

> KEY POINT † **Adrenal crisis management**: do not wait for results – draw cortisol/ACTH, then **hydrocortisone 100 mg IV bolus**, followed by 200 mg/24 h (infusion or 50 mg 6-hourly); rapid 0.9% saline (1 L in the first hour) with dextrose for hypoglycaemia; treat precipitant; taper to maintenance over 1–3 days. Fludrocortisone unnecessary while hydrocortisone >50 mg/day. Teach sick-day rules (double oral dose; injectable hydrocortisone) and a steroid alert card.

Surgical stress (on chronic steroids)	Example	Cover
Minor	Inguinal hernia, <1 h	Hydrocortisone 25 mg
Moderate	Joint replacement	Usual dose + hydrocortisone 50–75 mg intra-op
Major	Oesophagectomy, bypass	Usual dose + hydrocortisone 50 mg IV 8-hourly for 48–72 h

Glucocorticoid	Anti-inflammatory potency	Replacement dose (mg/day)	Half-life (h)
Hydrocortisone	1	20	8–12
Prednisone / prednisolone	4	5	18–36 (prednisone)
Dexamethasone	25–50	0.5	36–54

13 · CONGENITAL ADRENAL HYPERPLASIA (CAH)

- Autosomal recessive; **21-hydroxylase deficiency >95%**. Low cortisol → high ACTH → hyperplasia and androgen excess → *Ch 59*.
- Patients and carriers develop benign adenomas (not ACC); undiagnosed non-classic CAH may present as an adrenal mass in adults. Testicular adrenal rests can enlarge and mimic a testis tumour – consider before orchidectomy. No routine CAH screening of incidentalomas.

14 · LANDMARK TRIALS

Trial	Design	Result	Take-home
SPARTACUS	RCT: CT- vs AVS-guided PA treatment	Same antihypertensive intensity at 1 y; CT and AVS discordant in half; persistent aldosteronism 20% (CT) vs 10% (AVS)	AVS remains standard for biochemical cure
FIRM-ACT	RCT: mitotane + EDP vs mitotane + streptozotocin, advanced ACC	Better response and progression-free survival; overall survival similar	M-EDP first-line systemic therapy
ADJUVO	RCT: adjuvant mitotane vs observation, low/intermediate-risk R0 stage I–III ACC	No significant difference in recurrence	No routine adjuvant mitotane in low-risk disease

15 · MCQ ONE-LINERS

1. Unenhanced CT ≤10 HU in adrenal mass = lipid-rich adenoma.
2. Adenoma on 15-min washout: APW >60%, RPW >40%.
3. Only adrenal lesion exempt from the 4-cm resection rule: myelolipoma.
4. Every incidentaloma: 1-mg DST plus metanephrines; ARR if hypertensive or hypokalaemic.
5. 1-mg DST cortisol ≤1.8 µg/dL (50 nmol/L) excludes autonomous cortisol secretion.
6. Oral contraceptives cause false-positive DST by raising cortisol-binding globulin.
7. Most sensitive test for phaeochromocytoma: plasma free metanephrines (supine).
8. β-blockers cause false-positive ARR; ACE inhibitors/ARBs false-negative.
9. Commonest PA subtype: bilateral idiopathic hyperplasia (≈60%).
0. FH-I: CYP11B1/CYP11B2 chimeric gene, treated with glucocorticoid.
1. AVS may be skipped: age <40, unilateral adenoma >1 cm, normal contralateral gland.
2. AVS lateralisation ratio >2–4:1 (cortisol-corrected aldosterone).
13. Gold standard pituitary vs ectopic ACTH: inferior petrosal sinus sampling with CRH.
14. Commonest ectopic ACTH source: bronchial carcinoid.
15. Paradoxical cortisol rise after dexamethasone: PPNAD (Carney complex).
16. Nelson syndrome follows bilateral adrenalectomy for Cushing disease.
17. PPGL gene most associated with malignancy: SDHB.
18. SDHD shows maternal imprinting (disease when inherited from father).
19. VHL phaeochromocytoma secretes normetanephrine, not metanephrine.
20. FDG PET poor in MEN-2; ^{123}I-MIBG best there.
21. Never start β-blocker before α-blockade in phaeochromocytoma.
22. Extrapyramidal side effects: metyrosine, not phenoxybenzamine.
23. Malignant phaeochromocytoma is defined only by metastases.
24. Weiss criteria: ≥3 of 9 = adrenocortical carcinoma.
25. ENSAT stage IV ACC = distant metastasis only.
26. Primary adrenal insufficiency alone causes hyperkalaemia and hyperpigmentation.

147 Surgery of the Adrenal Glands

Reading time ≈ 18 min (slow pace)

1 · SURGICAL ANATOMY (ESSENTIALS)

- Three arteries, **one vein** — the operation is organised around the vein. Cortex yellow-orange, firm; medulla not seen.
- **Right**: pyramidal, behind bare area of liver (fused in ~10%), inferior vena cava (IVC) medial, duodenum anteromedial. Vein **<1 cm**, into **posterior IVC at 45°**; variants into right hepatic or right renal vein.
- **Left** → *Ch 145*: crescentic, more mobile; pancreatic body and splenic vessels anterior. Longer vein runs down, joined by inferior phrenic vein, into **left renal vein**.

2 · INDICATIONS

Two gates: **function** (operate whatever the size) and **malignant potential** (size, growth, imaging).

Category	Indication
Functional	Aldosteronoma (Conn); cortisol adenoma (Cushing syndrome); Cushing disease after failed trans-sphenoidal surgery; phaeochromocytoma (PCC); virilising/feminising tumour
Non-functional	Adrenocortical carcinoma (ACC); symptomatic cyst/myelolipoma; solitary metastasis
Size	**≥4 cm**; >2 cm growing ≥1 cm in 12 months
Imaging	Non-contrast CT (computed tomography) **>10 HU** (Hounsfield units); washout <50% with delayed >35 HU; chemical-shift MRI (magnetic resonance imaging) not adenoma; fluorodeoxyglucose positron emission tomography (FDG-PET) uptake ≥ liver

> **KEY POINT** Trap: bilateral hyperplasia causing aldosteronism is **not** surgical — hence adrenal venous sampling (AVS) before adrenalectomy for aldosteronism.

Incidentaloma work-up

- Mass ≥1 cm found incidentally (~5% of scans). ACC ~2%; autonomous secretion up to 20%. **Biochemistry before imaging.**
- **Cortisol** (all): 1-mg overnight dexamethasone suppression test (DST) — never spot cortisol.
- **Catecholamines** (all): plasma free or 24-h urinary metanephrines (continuous production); >2× upper limit significant.
- **Aldosterone** (if hypertensive): aldosterone-to-renin ratio (ARR) >20 with aldosterone >20 ng/dL → salt-loading confirmation → AVS. Up to 40% normokalaemic.
- **Sex steroids** if ≥4 cm or virilisation: dehydroepiandrosterone (DHEA) etc.
- Inert, ≤4 cm, ≤10 HU, homogeneous → repeat CT at 12 months. Myelolipoma (≤ −20 HU fat): operate only if **>7 cm**, symptomatic (haemorrhage) or atypical; still exclude PCC.

3 · CONTRAINDICATIONS

- Correctable states — operate **after** correction: uncontrolled hypertension, coagulopathy, bacterial peritonitis (transperitoneal route; insufflation drives bacteraemia). Low-dose aspirin may continue.
- Pregnancy (relative): defer elective surgery; necessary surgery (e.g. PCC) not delayed regardless of trimester.
- Prior nephrectomy may have removed the ipsilateral adrenal — confirm remaining tissue before contralateral adrenalectomy.

4 · PRE-OPERATIVE PREPARATION BY PATHOLOGY

Pathology	Preparation	Why
PCC	α-blockade **2 weeks** (phenoxybenzamine preferred; doxazosin, calcium-channel blocker, metyrosine alternatives) → β-blocker **only after α**, for tachycardia → volume expansion (fluids, high-salt diet)	Unopposed β-blockade removes β2 vasodilatation → hypertensive crisis. Chronically constricted, contracted patients collapse once vein tied.
Cushing	†Peri-operative hydrocortisone (e.g. 100 mg at induction), tapered over months; thromboprophylaxis	Contralateral gland suppressed → adrenal crisis. Steroid cover dropped early mortality to zero.
Aldosteronoma	†Correct K^+ (spironolactone/eplerenone, supplements); control blood pressure (BP)	†Hypokalaemia → arrhythmia under anaesthesia

- All PCC: electrocardiogram, echocardiogram (catecholamine cardiomyopathy); joint planning with anaesthesia.

5 · APPROACHES

	Open	Minimally invasive surgery (MIS)
Transperitoneal	Midline, subcostal, chevron	Laparoscopic, hand-assisted, robotic, single-site
Retroperitoneal	Flank, thoracoabdominal, posterior lumbodorsal	Laparoscopic, robotic, single-site

- **MIS is standard** since 1992 (first laparoscopic adrenalectomy, Gagner). **Lateral transperitoneal**: familiar anatomy, large working space.
- **Retroperitoneoscopic** (lateral or **prone** posterior): avoids bowel — preferred after prior abdominal surgery or with peritonitis; robot suits the small space. Bilateral small tumours prone without repositioning.
- **Open — when**: suspected/known **ACC**, large tumours, invasion of IVC/organs. Thoracoabdominal reserved for huge tumours or IVC extension above liver.

6 · OPERATIVE STEPS

Transperitoneal right

ALGORITHM
Elevate liver (no capsular tears) → Kocher (duodenum medial) → expose IVC → up from renal vein insertion (inferior limit) → find short adrenal vein → **warn anaesthesia**, clip/seal early → superior and lateral dissection → psoas (posterior limit) → bag, intact.

Transperitoneal left

ALGORITHM
Reflect colon at line of Toldt, outside Gerota → mobilise spleen/pancreas up → follow left renal vein (or gonadal vein) to adrenal vein → clip after warning → medial dissection, protect pancreas and splenic artery → divide ganglion nexus superomedially → bag.

Retroperitoneoscopic

- 1-cm incision below 12th rib tip → finger then balloon dissection → two ports 8 cm apart.
- Along psoas: **renal artery met first** (posterior to vein), unlike renal vein transperitoneally. Left vein anterocephalad to renal artery. Right vein enters posterolateral IVC above hilum — short, wide, avulsion → major bleed.
- Keep adrenal–kidney attachment last as a retraction handle (also in open).

> **KEY POINT** Why ligate the vein early in PCC? Limits catecholamine surge during handling; but hypotension follows — tell anaesthesia first.

7 · SPECIAL PATHOLOGIES

- **PCC intra-op**: arterial line; surge → nitroprusside, nicardipine, nitroglycerin, phentolamine, pause handling; tachyarrhythmia → esmolol, lidocaine.
- **ACC**: high-volume surgeon (>6, ideally ≥20/year); **open, intact, en bloc** with involved organs — capsular rupture is disastrous.
- **Metastasis**: lung and kidney ≈ two-thirds (breast uncommon). Operate when adrenal is the only disease; renal origin, <5 cm do best.
- **Congenital adrenal hyperplasia (CAH)**: bilateral adrenalectomy avoids iatrogenic Cushing from supraphysiological steroids; risks adrenal crisis, testicular rest tumours.
- **Ablation** (radiofrequency best, >50–60 °C coagulative necrosis): avoid for PCC or ACC outside trials.

8 · PARTIAL (CORTICAL-SPARING) ADRENALECTOMY

- **Indications**: bilateral/hereditary PCC, familial syndromes, prior contralateral adrenalectomy, children, Conn adenoma, cysts.
- **Principle**: arteries enter superiorly, medially, inferiorly — **lateral dissection safe**; preserve arteries; main vein may be divided (collateral outflow).
- Margins by intraoperative ultrasound; indocyanine green for remnant viability. Divide with clips, bipolar, sealing device or thin stapler; bag specimen.
- †Aim: retain about one-third of a gland to avoid lifelong steroids.

9 · COMPLICATIONS AND MANAGEMENT

Complication	Management
Haemorrhage (right vein avulsion, IVC, liver, spleen)	†Pressure, raise pneumoperitoneum, extra port; side-bite clamp and suture IVC; convert if uncontrolled
Hypotension after PCC vein ligation	Aggressive volume ± α-agonist (†noradrenaline/phenylephrine infusion); residual phenoxybenzamine prolongs it
Hypoglycaemia after PCC	†Insulin rebound once catecholamines fall; hourly glucose for 24–48 h, dextrose infusion
Adrenal insufficiency (Cushing, bilateral, partial)	†Hypotension, hyponatraemia, hyperkalaemia, fever → intravenous hydrocortisone 100 mg, saline; bilateral → lifelong gluco- + mineralocorticoid, sick-day rules
Hyperkalaemia after aldosteronoma	†Contralateral zona glomerulosa suppressed: stop K^+ and spironolactone; check K^+; occasionally fludrocortisone
Organ injury (pancreatic tail, spleen, pleura, bowel)	†Pancreatic leak → drain; pneumothorax → chest drain if significant; splenic tear → haemostatics/splenectomy
Flank bulge (open)	Protect subcostal neurovascular bundle during rib resection

10 · OUTCOMES AND FOLLOW-UP

- MIS: less pain, shorter stay than open; equivalent for benign disease.
- PCC: lifelong recurrence up to **16%** → annual blood pressure and metanephrines.
- †Aldosteronoma: hypokalaemia corrected in nearly all; hypertension cured in about half, improved in most.
- Cushing: metabolic syndrome resolves variably; insulin resistance may persist.
- ACC: advanced-stage 5-year survival **<15%** — multidisciplinary follow-up.

11 · MCQ ONE-LINERS

1. First laparoscopic adrenalectomy: Gagner, 1992, transperitoneal.
2. Steroid cover for Cushing was the first major safety advance.
3. Functional tumours excised regardless of size.
4. Non-functional incidentaloma ≥4 cm → adrenalectomy.
5. Non-contrast CT ≤10 HU → lipid-rich adenoma.
6. Best cortisol screen: 1-mg overnight DST.
7. Best PCC screen: plasma free or urinary metanephrines.
8. Up to 40% of primary aldosteronism is normokalaemic.
9. AVS distinguishes unilateral adenoma from bilateral hyperplasia.
0. Preferred α-blocker for PCC: phenoxybenzamine, ~2 weeks.
1. Never start β-blocker before α-blockade.
2. Right adrenal vein enters posterior IVC at 45°.
3. Left adrenal vein drains to left renal vein.
4. Kocher manoeuvre exposes IVC for right adrenalectomy.
5. Retroperitoneoscopy: renal artery met before vein.
6. Posterior limit of dissection: psoas.
7. Suspected ACC → open, intact, en bloc resection.
8. Commonest adrenal metastasis source: lung, then kidney.
9. Myelolipoma: operate if >7 cm or symptomatic.
0. Partial adrenalectomy: lateral dissection is safe.
1. In partial adrenalectomy main vein can be sacrificed.
2. Hypotension after PCC vein ligation → volume first.
3. PCC recurrence lifelong up to 16%.
4. Ablation of PCC or ACC discouraged outside trials.
5. Peritonitis contraindicates transperitoneal, not retroperitoneal, access.

140 Retroperitoneal Tumours

Reading time ≈ 34 min (slow pace)

1 · ANATOMY OF THE RETROPERITONEUM

- **Borders**: diaphragm (superior); inlet of true pelvis (inferior); peritoneal contents (anterior); posterior body wall (posterior); flank muscles (lateral).
- **Contents**: kidneys, renal pelvis, proximal/mid ureter; adrenals; infradiaphragmatic aorta and inferior vena cava (IVC), common iliacs, lumbar vessels; lymphatics, retroperitoneal (RP) fat, Gerota's fascia; sympathetic chain and postganglionic sympathetic/parasympathetic fibres.
- **Neighbours at risk** (tumours are expansile and distort or invade): duodenum (D2–D3), ascending/descending colon, pancreas, splenic hilum.

Region	Structures of greatest concern
Right upper RP	Kidney, ureter, adrenal; more importantly duodenum, IVC, posterior diaphragm, right liver; right colonic mesentery (loss of fat planes suggests involvement)
Left RP	Aorta, splenic hilum, pancreatic tail, diaphragm, left colonic mesentery
Both	Psoas and body wall — **depth of psoas invasion predicts nerve injury**

- Genitofemoral nerve injury → anterior/medial thigh paraesthesia. Femoral nerve injury → weak hip flexion and knee extension — functionally worse.
- Anticipate involved organs; book hepatobiliary, vascular, thoracic or general surgical help in advance.

> **KEY POINT** The operative problem is usually the neighbouring structure, not the tumour: RP tumours grow silently (resected sarcomas average **>15 cm**) and present with multi-organ involvement.

2 · CLASSIFICATION (DIFFERENTIAL DIAGNOSIS)

Group	Entities	First treatment
Germ cell / gonadal	Seminoma, nonseminoma (metastatic or extragonadal primary), sex cord stromal	Platinum chemotherapy (germ cell tumour, GCT); retroperitoneal lymph node dissection, RPLND (sex cord stromal)
Mesodermal — malignant	Liposarcoma, leiomyosarcoma, undifferentiated pleomorphic sarcoma (UPS), synovial, Ewing, desmoplastic small round cell tumour (DSRCT), unclassified	Surgery (most); chemotherapy (Ewing, DSRCT)
Mesodermal — benign/intermediate	Lipoma, leiomyoma, solitary fibrous tumour (SFT), perivascular epithelioid cell tumour (PEComa), desmoid, myelolipoma, gastrointestinal stromal tumour (GIST)	Excision / observation
Neurogenic (~10%)	Schwannoma, neurofibroma, malignant peripheral nerve sheath tumour (MPNST), ganglioneuroma, neuroblastoma, paraganglioma	Surgery
Haematological (~1/3 of primary RP tumours)	Hodgkin and non-Hodgkin lymphoma, plasmacytoma	Chemotherapy — biopsy only
Cystic	Lymphangioma, mucinous cystadenoma, cystic mesothelioma, cystic teratoma	Complete excision
Metastatic / unknown primary	Poorly differentiated carcinoma encasing great vessels	Histology-led

3 · CLINICAL EVALUATION

Presentation

- Usually incidental or painless mass; late symptoms from mass effect — venous obstruction (leg oedema), ureteric or bowel obstruction, nerve pain/paraesthesia.
- Sequence: **imaging → serology → tissue**.

Imaging

Modality	Role
Computed tomography (CT) chest/abdomen/pelvis — initial test of choice (National Comprehensive Cancer Network, NCCN)	Size, density, calcification, organ relations, border infiltration, nodal/distant spread. IV contrast: necrosis/cystic change; oral contrast: bowel involvement. **Negative Hounsfield units (HU, fat)** ≈ pathognomonic of liposarcoma, angiomyolipoma or myelolipoma. Cannot tell origin or malignancy.
Magnetic resonance imaging (MRI)	Vascular involvement when great vessels distorted; cystic/fatty components; contrast allergy or renal impairment
Positron emission tomography (PET)-CT	Selective: post-chemotherapy seminoma residual mass; grade/subtype of liposarcoma; lymphoma staging. False positives.

Serology

- Young man → **alpha-fetoprotein (AFP) and human chorionic gonadotrophin (hCG)** (± scrotal ultrasound); older man → prostate-specific antigen (PSA).
- Blood count (lymphoproliferative disease), renal/liver function; serum electrophoresis (myeloma M-spike). † Plasma metanephrines if paraganglioma is possible (para-aortic, organ of Zuckerkandl, hypertension).

Biopsy

- Consider fine-needle aspiration (FNA)/core biopsy in every RP mass; it earns its place where the result **changes the first treatment**: lymphoma (chemotherapy, never resection), GCT, Ewing, and high-grade sarcoma being considered for neoadjuvant therapy.
- **Core (14–16 G coaxial) over FNA**: more tissue, preserved architecture for grade and immunohistochemistry (IHC). † Image-guided via a **retroperitoneal (posterior) route**, avoiding transperitoneal passage and peritoneal seeding; the tract lies within the future resection field.
- **Avoid / unnecessary**: purely cystic lesions (rupture, poor yield); † suspected paraganglioma until catecholamine excess excluded/blocked (hypertensive crisis); raised AFP/hCG in a young man (markers diagnose GCT — start chemotherapy).
- Liposarcoma: imaging is characteristic and core biopsy is often non-diagnostic (scant atypical cells in fat).

4 · PATHOLOGY — MESENCHYMAL TUMOURS (SARCOMA)

- Soft-tissue sarcoma <1% of adult cancers; **15–20%** arise in the RP. Diagnosis needs integrated IHC, molecular and clinical data.

Entity	Key features	Treatment
Well-/de-differentiated liposarcoma (WD/DD; commonest, **~60%**)	Fat density; **MDM2/CDK4 (12q) amplification**, ring chromosomes; S100+	Radical, function-preserving compartmental resection
Leiomyosarcoma (second, **~20%**)	From vessel media (IVC), collecting-system muscle; no fat; SMA, desmin, h-caldesmon+	Negative margins + adjuvant therapy
UPS (formerly malignant fibrous histiocytoma)	5th–6th decade; heterogeneous, no fat, peripheral nodular enhancement; lineage markers negative	Wide resection

Entity	Key features	Treatment
Unclassified (up to 20%)	Diagnosis of exclusion; prior radiotherapy in a quarter	Screen MDM2 to exclude DD liposarcoma
Synovial	Calcification; TLE1; **t(X;18)**	Grade-dependent; chemosensitive
SFT (ex-haemangiopericytoma)	**NAB2-STAT6**; Doege-Potter hypoglycaemia (insulin-like growth factor excess)	En bloc, organ-sparing; radiosensitive
Ewing	Small round blue cell; EWSR1-FLI1 t(11;22); infiltrative	Chemotherapy-based multimodal
DSRCT	Young men; peritoneal implants, ascites; WT1-EWSR1 t(11;22)	Multimodal; universally poor
PEComa	Melanocytic (HMB-45) + muscle markers; TSC2	Excision; mammalian target of rapamycin (mTOR) inhibitors if unresectable
Desmoid	Nuclear β-catenin; APC (Gardner)/CTNNB1; never metastasises; recurs in up to half	Surgery or observation

Liposarcoma

- **WD vs DD** = presence of a higher-grade, non-lipogenic component. CT: fat-density mass; **internal higher-density nodules = de-differentiated areas** (both may coexist — target these if biopsied).
- MDM2/CDK4 fluorescence in situ hybridisation or IHC confirms in small samples.
- Behaviour: WD — slow, local recurrence up to 40%, rarely metastasises; DD — rapid, distant metastasis up to 20%, 5-year cancer-specific mortality ~30%. Mean time to recurrence ~**2 years** → lifelong surveillance.
- Unresectable/metastatic: anthracycline first line (modest); trabectedin (progression-free, not overall, survival benefit); pembrolizumab for high tumour mutational burden.
- **Pleomorphic liposarcoma**: MDM2/CDK4-negative; mimics carcinoma on core; early lung metastasis.

Leiomyosarcoma

- Presents by obstructing venous return, urine or bowel; in women many are gynaecological. Spectrum leiomyoma → STUMP (smooth muscle tumour of uncertain malignant potential) → leiomyosarcoma, judged on site, mitoses, atypia, necrosis.
- Needle biopsy underestimates aggressiveness; local and distant recurrence frequent → aggressive multimodal approach.

> **KEY POINT** Liposarcoma family sorts on MDM2: amplified (WD/DD) = local disease won or lost at the margin; absent (pleomorphic) = systemic disease. Infiltrative types (Ewing, DSRCT) need chemotherapy because no margin is achievable.

Grading and staging

- † **FNCLCC grade** (Fédération Nationale des Centres de Lutte Contre le Cancer) — sum of differentiation (1–3) + mitotic count (1–3 per 10 high-power fields: 0–9, 10–19, ≥20) + necrosis (0 none, 1 <50%, 2 ≥50%): **2–3 = G1, 4–5 = G2, 6–8 = G3**. Grade and histology (not size) are the main predictors of distant spread.

5 · NEUROGENIC TUMOURS

- Arise from sympathetic ganglia (behind great vessels), adrenals, organ of Zuckerkandl; mostly benign, prognosis better than sarcoma.

Tumour	Key features	Management
Schwannoma	Paravertebral; encapsulated, homogeneous, cystic change/calcification when large; S100+; slow	Complete local resection — curative (often enucleable)
Neurofibroma	Neurofibromatosis type 1 (NF-1) or sporadic; malignant change up to 20%	Surgical
MPNST	<10% of RP sarcoma; half NF-1; prior radiation; patchy S100, H3K27me3 loss; spreads along nerve roots	Resect with **parent nerve** (deficit inevitable); 5-year survival ~70% R0 vs ~20% R1
Ganglioneuroma	3rd–4th decade; calcification up to 30%	Excision — excellent
Ganglioneuroblastoma / neuroblastoma	MYCN amplification adverse; MIBG (metaiodobenzylguanidine) staging; often metastatic	Paediatric-regimen chemotherapy ± surgery
Paraganglioma	Commonest site organ of Zuckerkandl (anterior aorta below inferior mesenteric artery, IMA); T2-bright; secretory in ~40%; metastases 20–50% (histology poor predictor); NF-1, multiple endocrine neoplasia 2A, SDHB	Surgery (after α-blockade†) → *Ch 145*

6 · GERM CELL, HAEMATOLOGICAL, CYSTIC AND UNKNOWN PRIMARY

- **GCT**: RP mass in a young man is a GCT until proven otherwise. Extragonadal RP GCT (3–5%) is often a **burnt-out testicular primary** (scar/teratoma at orchidectomy). Management: platinum chemotherapy first → resect all nonseminomatous residual masses (teratoma is chemoresistant and marker-silent). Isochromosome 12p is the marker. → *Ch 79–80*
- **Sex cord stromal** (Leydig, Sertoli, granulosa): ~10% metastasise (RP first); chemo/radioresistant and surveillance deaths high → early RPLND in clinically node-negative high-risk primaries (age, >4–5 cm, necrosis, mitoses, atypia, lymphovascular invasion).
- **Lymphoma**: homogeneous, confluent masses along nodal chains; biopsy, never resect. **Plasmacytoma**: M-spike, bone marrow, CD138; wide excision/RT/chemotherapy for isolated disease.
- **Cystic lymphangioma**: thin, multiseptated, crosses midline; male children. Mucinous cystadenoma and cystic mesothelioma: women. All → **complete excision** (aspiration/marsupialisation abandoned — recurrence).
- **Cancer of unknown primary**: primary found in <20%. Midline/i(12p)+ → platinum chemotherapy; single small deposit → excision/focused radiotherapy.

> **KEY POINT** Histology decides first treatment more than anatomy: exclude chemotherapy-first entities (GCT, lymphoma, Ewing, midline unknown primary) before planning a sarcoma resection.

7 · SURGERY FOR RP SARCOMA — PRINCIPLES

- **Complete en bloc (R0) resection at the first operation** in an experienced centre is the backbone of cure; † the first operation is the best chance.
- **Compartmental / multivisceral resection**: liposarcoma margins are ill-defined in fat, so tumour is removed en bloc with adherent, uninvolved neighbours (kidney, colon, psoas fascia) — reduces local recurrence. Preserve function-critical structures (duodenum, pancreas head, major vessels, femoral nerve) unless directly invaded. † **Never enucleate** a sarcoma (pseudocapsule contains tumour).
- Nephrectomy is the commonest auxiliary procedure: up to **55%** in primary RP sarcoma; renal function falls but rarely needs dialysis — † confirm contralateral function (split renogram) beforehand.
- About 5% need major lumbar nerve resection (femoral, obturator) → knee brace may be required.

Preoperative preparation

- CT within 6 weeks, reviewed again immediately before surgery; chest CT if lung disease. No bowel preparation; cross-match extra blood.
- GCT: markers within 7–10 days; sperm banking. Prior **bleomycin** → pulmonary function tests, low FiO_2 (fraction of inspired oxygen) and restrictive fluids (prevents acute respiratory distress syndrome).
- Complete IVC thrombosis on imaging simplifies surgery (IVC resected, collaterals present); partial occlusion may need PTFE (polytetrafluoroethylene) or allograft reconstruction if no bowel contamination.

Exposure and technique

- Midline laparotomy; falciform divided (prevents liver retraction injury); orogastric tube (nasogastric only for duodenal work).
- Right/central masses: incise mesenteric root from caecum, extending along right line of Toldt to foramen of Winslow (bowel on chest); left masses: left line of Toldt ± divide inferior mesenteric vein.
- **Split-and-roll**: split each great vessel at 12 o'clock (aorta from just below left renal vein), roll tissue off its sides, ligate lumbar vessels (arteries predictable, veins not), protect sympathetic trunks and genitofemoral nerve, ligate retrocrural lymphatics (chylous ascites). IMA preserved on right-sided, divided in bilateral dissection.

Vascular resection

- **Aorta**: always reconstructed (tube/aortobi-iliac graft) after heparin and cross-clamp — obligatory flow.
- **IVC**: gradually occluded cava has collaterals → ligation tolerated. In sarcoma, caval resection ± graft for R0; grafts thrombose in 10–40% → anticoagulate ≥3 months.

8 · RPLND IN GERM CELL TUMOURS (ESSENTIALS)

- Lymphatic spread is side-specific: **right testis → interaortocaval/precaval; left → left para-aortic**; contralateral and suprahilar spread rises with bulk.
- Suprahilar/retrocrural and interiliac dissection omitted from standard templates. Primary RPLND: modified unilateral vs full bilateral infrahilar debated — survival ~100% either way.
- **Post-chemotherapy RPLND (PC-RPLND)**: all residual disease + full bilateral infrahilar template (standard). Modified unilateral only if: lesion **≤5 cm** in primary landing zone before and after chemotherapy, normal markers, IGCCCG (International Germ Cell Cancer Collaborative Group) good/intermediate risk.
- **Nerve-sparing**: postganglionic fibres L1–L4 converge on the superior hypogastric plexus below the IMA; identify prospectively, loop, dissect packets around them → antegrade ejaculation **90–100%**. Temporary anejaculation = neuropraxia.

9 · COMPLICATIONS AND MANAGEMENT

Complication	Cause / prevention	Management
Chylous ascites (2–7% post-chemotherapy)	Suprahilar dissection (cisterna chyli L1–L2, retrocrural), IVC resection, long surgery → ligate retrocrural lymphatics	Graduated (below)
Lymphocele	Unligated lymphatics	Percutaneous drain; leave drain if infected
Pancreatic leak	Left suprahilar work injures tail	Check drain amylase/lipase
Anejaculation	Bilateral/caudal-to-IMA dissection	Nerve-sparing; sperm banking

Complication	Cause / prevention	Management
Venous thromboembolism	~1 in 5 after caval reconstruction	Prophylaxis, anticoagulation
Nerve injury	Femoral neuropraxia from retractors; paraplegia after combined mediastinal + RP resection	Careful positioning; brace

ALGORITHM

Chylous ascites: paracentesis + low-fat medium-chain triglyceride (MCT) diet ± octreotide → reaccumulation: indwelling drain → nil by mouth + total parenteral nutrition → persistent >100 mL/day: peritoneovenous (LeVeen) shunt or surgical ligation (last resort) → continue MCT diet 1–3 months after resolution.

10 · ADJUVANT AND NEOADJUVANT THERAPY

Radiotherapy (RT)

- Aims to reduce local recurrence. Late bowel toxicity (50–60 Gy) is mostly with post-operative RT; **pre-operative RT** is better tolerated †(tumour displaces bowel, defined target).
- **STRASS**: pre-operative RT did not improve abdominal recurrence-free survival → **not standard**; † exploratory benefit in liposarcoma (especially low-grade) keeps it an option in selected cases. intraoperative radiotherapy (IORT) and brachytherapy investigational.

Chemotherapy

- Liposarcoma (WD) and leiomyosarcoma largely chemoresistant; pooled series show no benefit. Responsive: myxoid liposarcoma, synovial, Ewing — **standard neoadjuvant only for Ewing**.
- Resectability doubtful or regional/distant spread → induction multimodal therapy; STRASS II (neoadjuvant doxorubicin + ifosfamide/dacarbazine) ongoing.

11 · RECURRENCE AND FOLLOW-UP

- † Local recurrence is the main cause of death in RP liposarcoma; leiomyosarcoma fails distantly (lung, liver). Recurrence ~2 years on average but can be late.
- † CT abdomen/pelvis (and chest) every 3–6 months for 2–3 years, then 6-monthly to 5 years, then annually — lifelong for liposarcoma.
- Recurrent disease: re-resection if resectable and slow-growing (WD liposarcoma), each subsequent R0 less likely; otherwise systemic therapy or trials.

12 · LANDMARK TRIALS

Trial	Design	Result	Take-home
European Organisation for Research and Treatment of Cancer (EORTC)-62092 STRASS	Phase III randomised, 266 patients: pre-op RT + surgery vs surgery	No abdominal recurrence-free survival difference (hazard ratio 1.01); more complications	Pre-op RT not standard for RP sarcoma
STRASS II (ongoing)	Neoadjuvant chemotherapy + surgery vs surgery	Awaited	Neoadjuvant chemotherapy remains unproven

13 · MCQ ONE-LINERS

1. Commonest primary RP sarcoma: liposarcoma (~60%); second: leiomyosarcoma (~20%).
2. Negative HU fat on CT: liposarcoma, angiomyolipoma, myelolipoma.
3. WD/DD liposarcoma: MDM2/CDK4 amplification on 12q; pleomorphic liposarcoma is MDM2-negative.
4. Dense nodule within a fatty RP mass suggests de-differentiation.
5. SFT: NAB2-STAT6 fusion; Doege-Potter syndrome = IGF-mediated hypoglycaemia.
6. Synovial sarcoma: t(X;18), TLE1, calcification.

7. Ewing and DSRCT both t(11;22) EWSR1 — partners FLI1 vs WT1.
8. Lymphoma in RP: biopsy, then chemotherapy — never resect.
9. Young man with RP mass: AFP and hCG first.
0. Extragonadal RP GCT is often a burnt-out testicular primary.
1. Isochromosome 12p marks GCT, including unknown-primary midline tumours.
2. Commonest extra-adrenal paraganglioma site: organ of Zuckerkandl.
3. MPNST: H3K27me3 loss; half in NF-1; parent nerve must be sacrificed.
4. Desmoids never metastasise but recur locally; nuclear β-catenin.
15. PEComa unresectable: mTOR inhibitors.
16. FNCLCC grade: differentiation + mitoses + necrosis; score 6–8 = grade 3.
17. Cystic lymphangioma: complete excision; aspiration/marsupialisation recur.
18. Aorta always reconstructed; chronically occluded IVC can be ligated.
19. Chylous ascites first line: paracentesis + MCT diet ± octreotide.
20. Right testis drains interaortocaval; left drains para-aortic.
21. Unilateral PC-RPLND only if mass ≤5 cm in landing zone, normal markers, good/intermediate risk.
22. Only Ewing sarcoma has standard neoadjuvant chemotherapy among RP sarcomas.

★ Putting it together: adrenal and retroperitoneal decisions

Reading time ≈ 5 min (slow pace) · Integrates Ch 140, 145–147 · use these as the skeleton of long-essay answers

A · ADRENAL INCIDENTALOMA: TWO QUESTIONS

ALGORITHM

1. Is it malignant? Unenhanced CT ≤10 Hounsfield units (HU), or signal drop-out on chemical-shift MRI → lipid-rich adenoma. If >10 HU → washout CT (absolute washout >60% or relative >40% = adenoma). Size >4 cm, suspicious imaging or growth >1 cm → consider resection (→ *Ch 146*).

2. Is it functioning? Test everyone: 1 mg overnight dexamethasone suppression test (cortisol), plasma or urine metanephrines (phaeochromocytoma), and aldosterone–renin ratio if hypertensive or hypokalaemic.

Silent, <4 cm, benign imaging → no further follow-up needed.

B · FUNCTIONAL ADRENAL TUMOURS COMPARED

	Primary aldosteronism	Cushing syndrome	Phaeochromocytoma
Screen	Aldosterone–renin ratio	1 mg dexamethasone suppression, late-night salivary cortisol, 24-h urinary free cortisol	Plasma free or urine fractionated metanephrines
Next step	Confirmatory test → CT → adrenal vein sampling to lateralise	ACTH: suppressed = adrenal; normal/high = pituitary or ectopic	CT/MRI, functional imaging; genetic testing for all
Before surgery	Correct potassium (spironolactone/eplerenone)	Perioperative steroid cover	α-blockade first, then β-blocker; fluids and salt
After surgery	Stop potassium and spironolactone; watch for hyperkalaemia	Steroid replacement until the axis recovers (months)	Watch for hypotension and hypoglycaemia
Medical alternative	Mineralocorticoid antagonist (bilateral disease)	Steroidogenesis inhibitors	Long-term α-blockade if inoperable

C · ADRENALECTOMY: WHICH APPROACH?

KEY POINT Laparoscopic (transperitoneal lateral or posterior retroperitoneoscopic) is standard for benign tumours. Choose open surgery for large suspected adrenocortical carcinoma or local invasion, because capsular rupture causes tumour spill and recurrence. Right side: the short adrenal vein drains straight into the inferior vena cava, so control it carefully. Partial adrenalectomy for bilateral or hereditary disease preserves function (→ *Ch 147*).

D · RETROPERITONEAL MASS

ALGORITHM

Image (CT/MRI) → **exclude** paraganglioma (metanephrines), germ cell tumour (testis exam, markers) and lymphoma before biopsy.

Biopsy by core needle via a retroperitoneal route when the diagnosis would change treatment.

Sarcoma: complete en bloc resection with negative margins at the first operation (multivisceral if needed) in a specialist centre; never enucleate. Lifelong CT follow-up, especially for liposarcoma (→ *Ch 140*).